



SNIP: Adaptive Mixed Precision for Subbyte LLM Training

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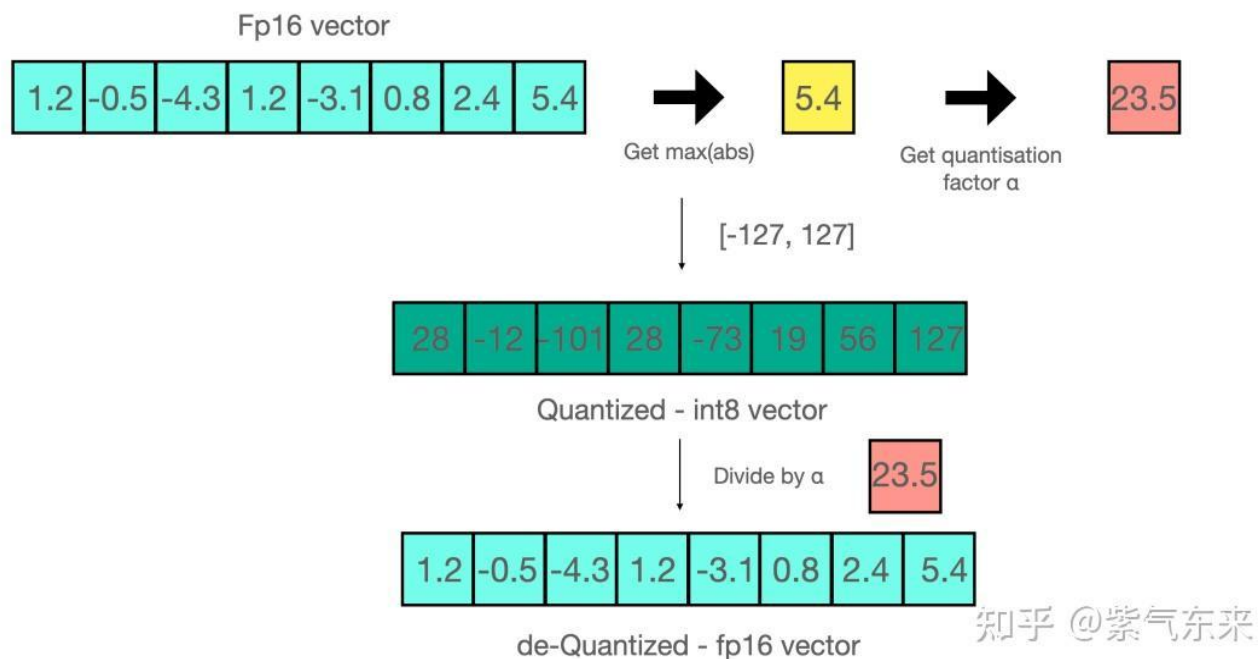
**Presenter: YiFan Hu
2026.8.26**

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- Background & Motivation
- Methodology
- Evaluation
- Thinking

Background & Motivation

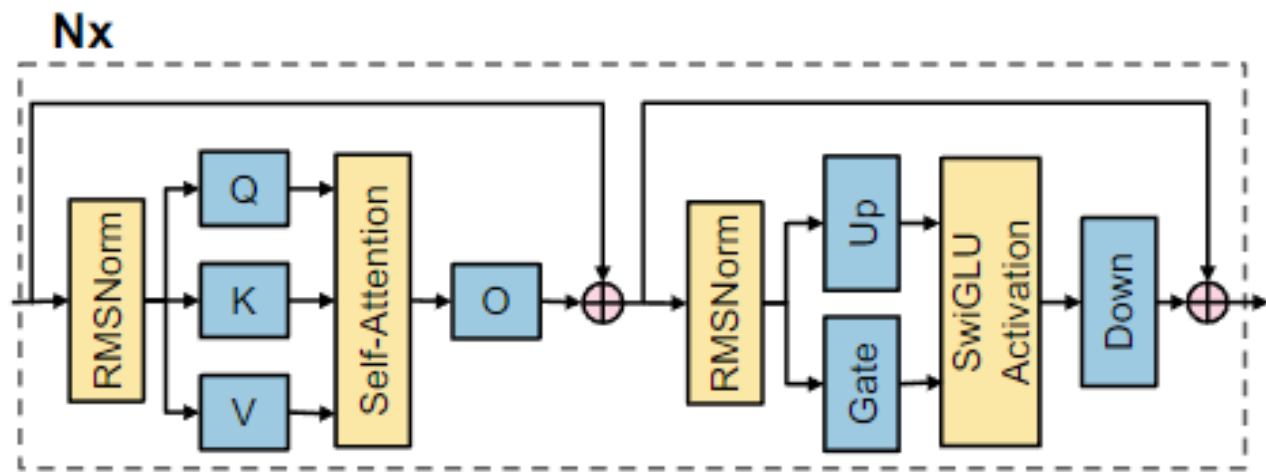
量化训练与推理:



显卡在以FP8格式计算矩阵乘法时能达到FP16计算的两倍吞吐量，FP4达到4倍

Background & Motivation

- 大模型训练成本高：使用80亿个参数训练llama需要在H100 GPU上运行146万GPU小时
- 硬件趋势：H100 已支持 FP8 张量核（相对 FP16 $\approx 2\times$ 吞吐）；Blackwell（B200）支持 FP6/FP4
- 混合精度：一种训练框架，精细化配置每一层的精度，实现分层量化



Background & Motivation

问题:

- 使用混合精度于训练过程中, 如何保证最终训练的loss结果几乎不下降
- 混合精度分配问题是一个NP难问题, 每一层的精度选择是离散的, 假设模型有 L 层, 每一层可以从 K 种精度中选择一种, 最终配置数量超穷举

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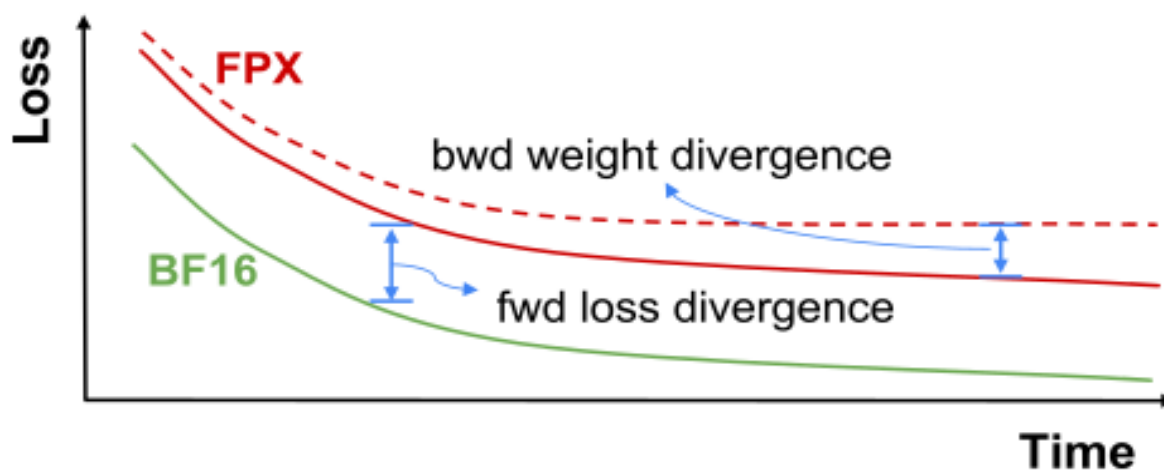
Methodology

解决训练精度问题：

- 损失误差：当量化增加训练损失时，前向传递会出现损失偏差
- 权重误差：当量化误差使参数更新失真，后向传递会发生权重偏差

解决NP难问题：

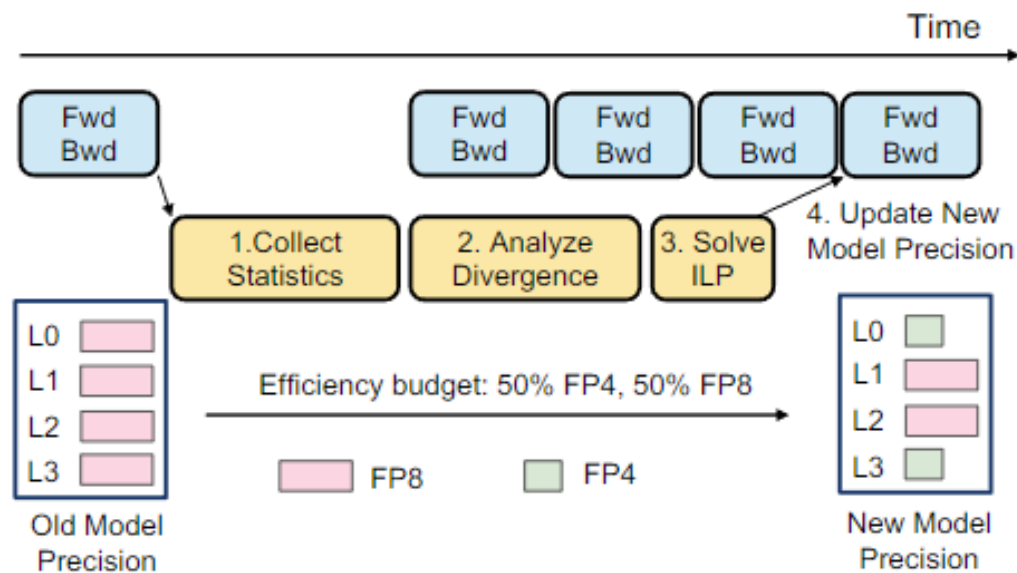
- 将问题建模为ILP问题，并在实际模型规模下使用现代ILP求解器高效求解



Methodology

生成分层量化方案(FPX)方案，主要包括两个步骤

- 收集统计数据和分析的偏差
- 确定最佳逐层量化方案



Methodology

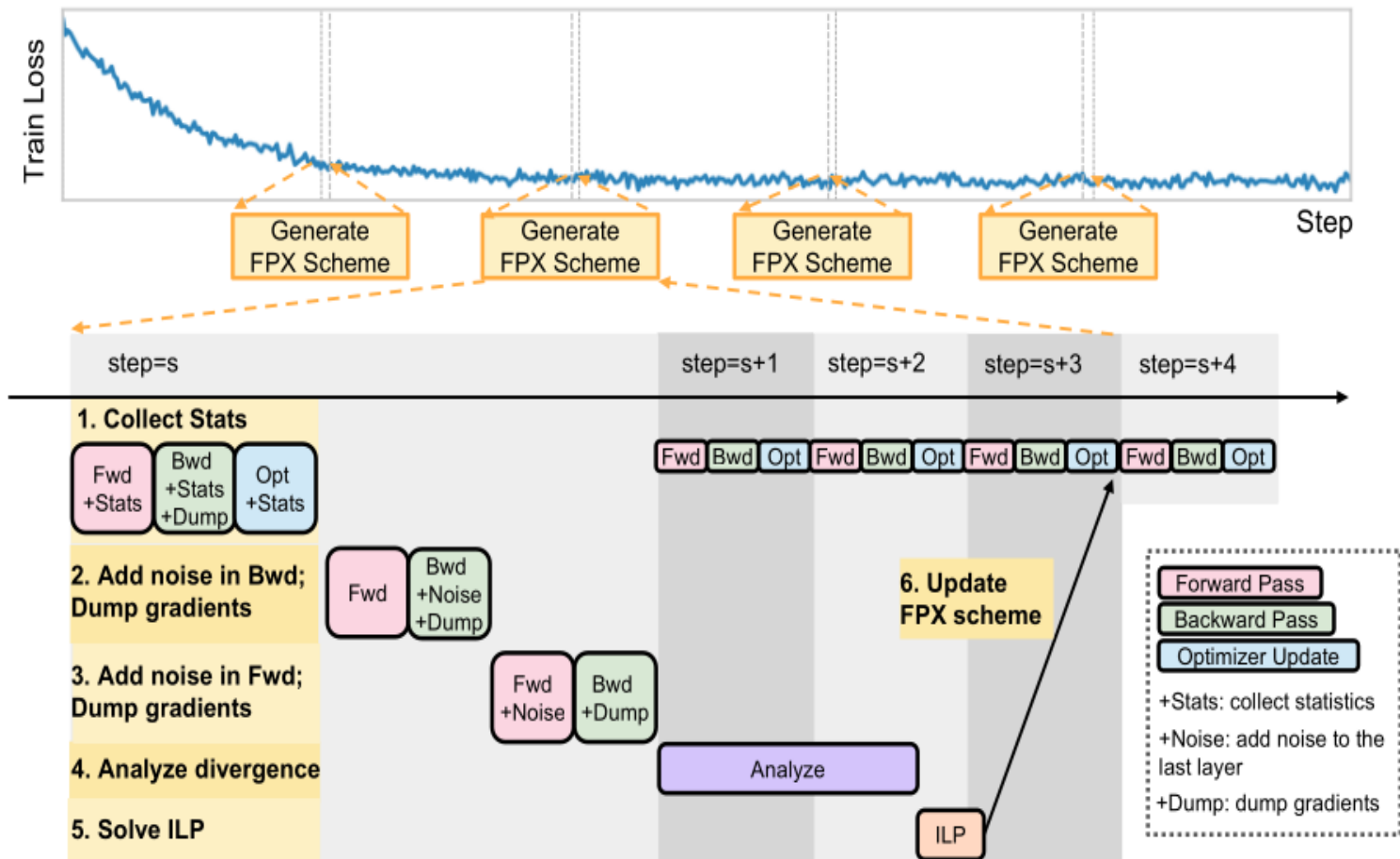
1 收集数据

$$\|X_l\|_F, \|W_l\|_F, \|\nabla_{X_l} L\|_F, \|\nabla_{W_l} L\|_F$$

$$\|\delta X_l\|_F, \|\delta W_l\|_F, \|\delta g_l\|_F$$

完整的梯度张量，也就是dump

优化器中的一阶矩和二阶矩



Methodology

2-3 增加噪声

损失 L

→ 第L层 (添加噪声 ϵ)

→ 第L-1层

→ ...

增加噪声后重新做一次反向传播, 得到 g'_l

$$\Delta g_l^{\text{measured}} = g'_l - g_l$$

$$\frac{\partial g_l}{\partial(\text{被扰动的东西})} \approx \frac{\Delta g_l^{\text{measured}}}{\epsilon}$$

$$\Delta g_l \approx \frac{\partial g_l}{\partial(\nabla Y_j)} \cdot \delta(\nabla Y_j)$$

2. Add noise in Bwd;
Dump gradients



3. Add noise in Fwd;
Dump gradients



输出变化 $\approx$ 敏感度 $\times$ 扰动大小

$$\frac{\partial g_l}{\partial(\nabla Y_j)} = \frac{\partial}{\partial(\nabla Y_j)} \left(\frac{\partial L}{\partial W_l} \right)$$

$$\frac{\partial g_l}{\partial X_j} = \frac{\partial}{\partial X_j} \left(\frac{\partial L}{\partial W_l} \right)$$

Methodology

4.1 损失误差

如果第L层量化，最终loss增大多少

输出变化 $\approx$ 敏感度 $\times$ 扰动大小

$$\|\nabla_{X_l} L\|_F, \|\nabla_{W_l} L\|_F \quad \|\delta X_l\|_F, \|\delta W_l\|_F$$

根据一阶泰勒展开：

$$L(X + \delta X) - L(X) \approx \frac{\partial L}{\partial X} \cdot \delta X$$

$$\Delta L_X \approx \|\nabla_{X_l} L\|_F \cdot \frac{\|\delta X_l\|_F}{\sqrt{M_l K_l}}$$

Methodology

4.2 权重误差 梯度变化量 $\approx$ 梯度对扰动的敏感度 $\times$ 扰动大小

$$\Delta g_l \approx \frac{\partial g_l}{\partial (\nabla Y_j)} \cdot \delta(\nabla Y_j) \quad g_l(X_j + \delta X_j, W_j, \nabla Y_j) - g_l(X_j, W_j, \nabla Y_j) \approx \|\nabla_{X_j} g_l\|_F \cdot \frac{\|\delta X_j\|_F}{\sqrt{M_j K_j}}$$

$$W_t = (1 - \alpha\lambda)W_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad m_t = \beta_1 m_{t-1} + (1 - \beta_1)g_t \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2)g_t^2$$

$$W_t - W'_t = (1 - \alpha\lambda)(W_{t-1} - W'_{t-1}) - \alpha \left(\frac{\hat{m}'_t}{\sqrt{\hat{v}'_t} + \epsilon} - \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \right)$$

$$U(g) = \alpha \frac{\hat{m}}{\sqrt{\hat{v}} + \epsilon} \quad \nabla_g h(g) = \alpha \frac{\sqrt{1 - \beta^t}}{1 - \beta^t} \left[\frac{1 - \beta_1}{\sqrt{v_t} + \epsilon} - \frac{(1 - \beta_2)m_t g_t}{\sqrt{v_t}(\sqrt{v_t} + \epsilon)^2} \right]$$

$$U(g + \Delta g) - U(g) \approx \nabla_g U(g) \cdot \Delta g$$

Methodology

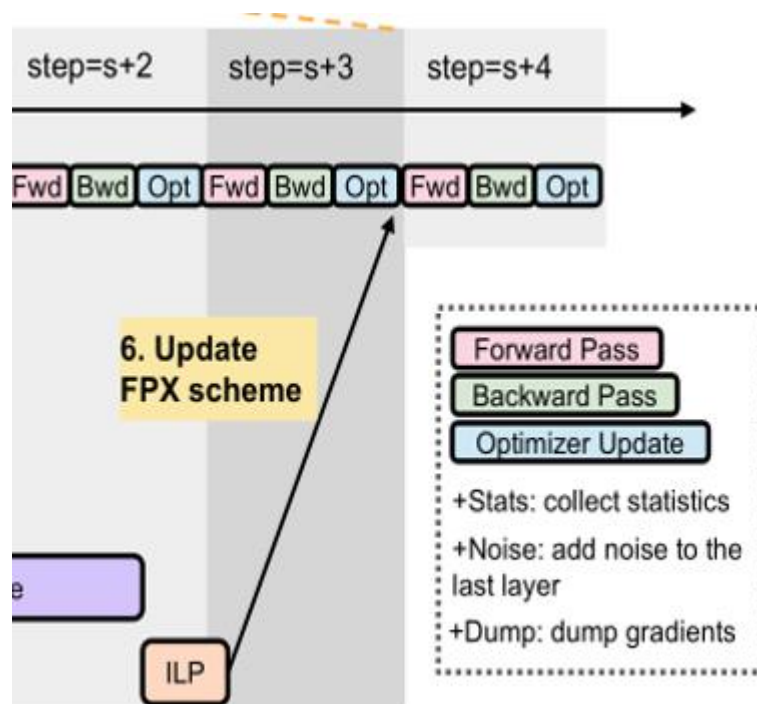
5-6. 解决ILP问题和更新

质量损失 $Q = \Delta L + \Delta W$

效率收益 $E = \text{FP4 FLOPs 占总 FLOPs 的比例}$

$$\begin{aligned} \{x_{i,j}^*\} &= \arg \min_{\{x_{i,j}\}} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} q_{i,j} x_{i,j} \\ \text{s.t. } &\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} e_{i,j} x_{i,j} \geq E_t \\ &\sum_{j=0}^{n-1} x_{i,j} = 1, \quad \forall i \in \{0, 1, \dots, m-1\} \\ &x_{i,j} \in \{0, 1\}, \quad \forall i, j \end{aligned}$$

$$\text{s.t. } \sum_{i=k*g}^{(k+1)*g-1} \sum_{j=0}^{n-1} e_{i,j} x_{i,j} \geq \frac{E_t}{K}, \forall k \in 0, 1, \dots, K$$



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Evaluation

Fraction of FP4 FLOPS	Quant schemes	Math		Aggregate	Commonsense					Average
		ARC_c	ARC_e	MMLU	BoolQ	HellaSwag	Obqa	PiQa	WinoGrande	
0	BF16	24.15	46.09	26.56	59.72	43.94	31.00	67.41	54.85	44.22
	FP8	23.98	45.71	26.61	60.15	44.04	31.20	67.46	54.46	44.20
25%	SNIP	24.06	45.75	26.73	60.28	43.84	31.20	67.25	54.78	44.24
	min-abs-err	24.06	45.75	26.72	60.55	43.99	30.20	67.57	53.99	44.10
	min-rel-err	24.49	45.71	26.22	60.52	43.72	30.60	67.68	54.30	44.16
	random0	23.21	39.44	25.32	59.42	35.58	28.40	63.76	51.54	40.83
	random1	23.98	29.08	25.69	39.39	28.22	23.80	54.08	49.49	34.22
	random2	24.06	44.65	26.14	61.25	43.07	30.00	67.30	53.75	43.78
	E-layer-type	27.82	26.43	22.95	37.83	26.01	25.60	48.97	50.04	33.21
50%	SNIP	24.06	45.41	27.04	60.43	43.79	30.40	67.57	54.54	44.16
	min-abs-err	25.60	26.98	25.41	38.04	26.50	25.80	51.25	50.04	33.70
	min-rel-err	27.56	28.16	24.39	37.83	26.64	26.40	52.34	51.22	34.32
	random0	26.62	27.65	23.09	37.83	27.26	24.60	51.96	49.64	33.58
	random1	28.41	27.53	24.79	39.36	27.88	26.20	51.63	49.57	34.42
	random2	24.15	44.61	25.76	61.25	42.74	30.40	66.92	53.04	43.61
	E-layer-id	26.37	26.56	24.80	37.83	27.19	25.20	49.46	48.70	33.26
	E-layer-type	26.71	26.98	22.94	37.83	26.47	24.20	50.22	49.57	33.11
75%	SNIP	24.40	46.04	26.68	60.76	43.46	30.80	67.41	54.14	44.21
	min-abs-err	27.22	26.52	24.66	37.83	26.40	25.40	50.92	49.25	33.53
	min-rel-err	25.94	26.18	24.28	37.83	26.50	25.80	51.09	50.67	33.54
	random0	27.56	27.15	25.30	37.83	26.21	26.20	51.14	49.88	33.91
	random1	28.67	27.57	24.75	37.83	25.99	24.60	51.41	50.20	33.88
	random2	27.39	27.36	25.74	37.83	26.56	26.00	50.76	49.33	33.87
80%	SNIP	24.23	46.00	26.83	61.07	43.16	31.80	67.36	53.67	44.27
85%	SNIP	25.60	26.64	26.50	37.83	26.55	25.80	52.99	50.20	34.01
100%	FP4	26.11	27.02	23.12	37.83	25.97	23.60	50.33	51.46	33.18

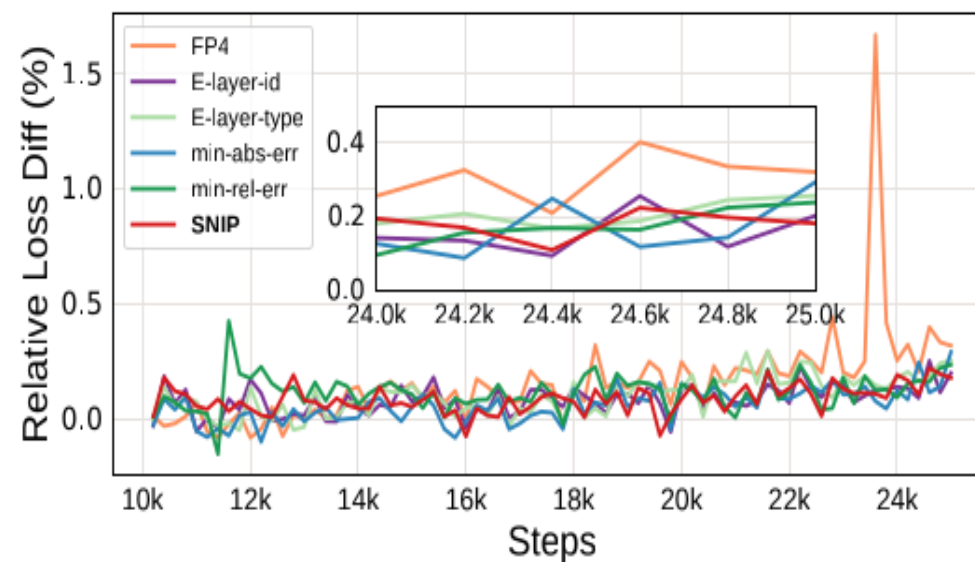
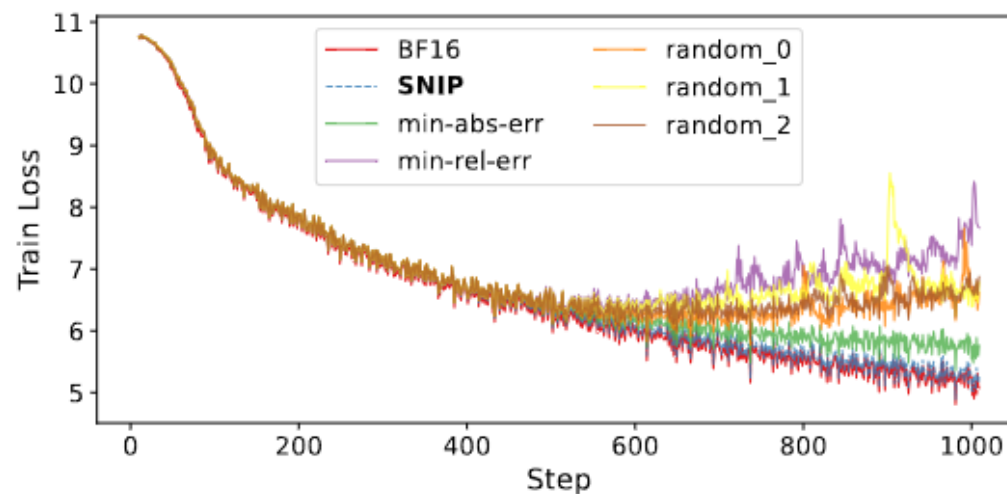
Evaluation

TinyLlama 1B(75% FP4):BF16和SNIP几乎重叠

OpenLlama 1B/3B/7B: 准确率高

Llama 70B: 50%FP4保持稳定

Model	TinyLlama 1B			OpenLlama 3B		OpenLlama 7B	
Checkpoint	5k	50k	240k	50k	100k	50k	100k
BF16	39.16	44.22	46.13	49.94	51.85	52.46	54.42
SNIP	39.15	44.21	45.78	49.72	51.72	52.43	54.45
min-abs-err	39.07	33.53	45.54	49.95	51.99	52.15	54.40
min-rel-err	33.11	33.54	45.57	50.19	51.93	52.42	54.34
random0	33.71	33.91	45.23	48.16	49.59	49.52	52.87
random1	33.24	33.88	32.86	49.16	51.04	49.05	51.84
random2	38.96	33.87	33.55	48.35	51.32	50.13	52.08



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Thinking

- **文章有什么问题，基于这篇 paper 还能做什么优化？**
 - 其效率选择为FP4的占比，而非实际的FP4/FP8的kernel时间，80%的FP4不能看作训练加速80%
 - 没有预测长期训练质量，应该考虑当前做的持续性量化将给最终的loss带来的变化
- **这篇 paper 的 idea 能不能应用在自己的工作上面？**
 - 从训练层转移到通信上，实现通信的压缩自适应
 - 在求解问题时以ILP作为初始求解框架



Q & A

Presenter: YiFan Hu
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